Appendix Experiments and Results

The subdirectories contain the experiment code and run logs for Tables 4 and 6–13. Results corresponding to Tables 1, 4, and 6–13 are presented below.

Table 1: Multivariate forecasting results. The look-back length L is set to 96. The best results are highlighted in **bold** and the second best are underlined.

Models		**CTIA-ODE (Ours)**		WaveTS-B		S-Mamba		TimeXer		iTransformer		PatchTST		DLinear		Crossformer		TimesNet		Autoformer	
Metric		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
Electricity	96	**0.140**	**0.233**	0.143	0.235	0.147	0.249	0.147	0.251	0.148	0.240	0.195	0.285	0.197	0.282	0.219	0.314	0.168	0.272	0.201	0.317
	192	**0.160**	**0.252**	0.164	0.255	0.167	0.270	0.163	0.266	0.162	0.253	0.199	0.289	0.196	0.285	0.231	0.322	0.184	0.289	0.222	0.334
	336	0.181	0.274	0.185	0.278	0.189	0.291	0.184	0.282	**0.178**	**0.269**	0.215	0.305	0.209	0.301	0.246	0.337	0.198	0.300	0.231	0.338
	720	**0.205**	**0.296**	0.211	0.302	0.232	0.327	0.220	0.312	0.225	0.317	0.256	0.337	0.245	0.333	0.280	0.363	0.220	0.320	0.254	0.361
	Avg	**0.172**	**0.264**	0.176	0.268	0.183	0.284	0.178	0.278	0.178	0.270	0.216	0.304	0.212	0.300	0.244	0.334	0.192	0.295	0.227	0.338
Exchange	96	0.093	0.215	0.098	0.219	0.089	0.211	0.101	0.226	0.118	0.253	**0.088**	**0.205**	**0.088**	0.218	0.256	0.367	0.107	0.234	0.197	0.323
	192	0.188	0.308	0.192	0.311	0.187	0.309	0.199	0.317	0.232	0.355	**0.176**	**0.299**	**0.176**	0.315	0.470	0.509	0.226	0.344	0.300	0.369
	336	0.325	0.415	0.342	0.428	0.338	0.423	0.334	0.424	0.434	0.489	**0.301**	**0.397**	0.313	0.427	1.268	0.883	0.367	0.448	0.509	0.524
	720	**0.788**	**0.668**	1.031	0.768	0.875	0.710	0.803	0.680	1.050	0.750	0.901	0.714	0.839	0.695	1.767	1.068	0.964	0.746	1.447	0.941
	Avg	**0.348**	**0.402**	0.415	0.431	0.372	0.413	0.359	0.411	0.459	0.463	0.367	0.404	0.354	0.414	0.940	0.707	0.416	0.443	0.613	0.539
Traffic	96	0.411	**0.253**	0.416	0.271	0.428	0.271	0.435	0.279	**0.395**	0.268	0.462	0.295	0.650	0.396	0.522	0.290	0.593	0.321	0.613	0.388
	192	0.425	**0.262**	0.431	0.278	0.442	0.280	0.457	0.288	**0.417**	0.276	0.466	0.296	0.598	0.370	0.530	0.293	0.617	0.336	0.616	0.382
	336	0.456	**0.270**	0.462	0.281	0.471	0.292	0.483	0.296	**0.433**	0.283	0.482	0.304	0.605	0.373	0.558	0.305	0.629	0.336	0.622	0.337
	720	0.474	**0.285**	0.481	0.302	0.495	0.308	0.522	0.316	**0.467**	0.302	0.514	0.322	0.645	0.394	0.589	0.328	0.640	0.350	0.660	0.408
	Avg	0.441	**0.268**	0.448	0.283	0.459	0.288	0.474	0.293	**0.428**	0.282	0.481	0.304	0.625	0.383	0.550	0.304	0.620	0.336	0.628	0.379
Weather	96	**0.153**	**0.191**	0.172	0.225	0.169	0.210	0.165	0.214	0.174	0.214	0.177	0.218	0.196	0.255	0.158	0.230	0.172	0.220	0.266	0.336
	192	0.208	**0.247**	0.215	0.261	0.214	0.253	0.211	0.253	0.221	0.254	0.225	0.259	0.237	0.296	**0.206**	0.277	0.219	0.261	0.307	0.367
	336	**0.267**	**0.288**	0.273	0.298	0.274	0.296	0.271	0.298	0.278	0.296	0.278	0.297	0.283	0.335	0.272	0.335	0.280	0.306	0.359	0.395
	720	0.351	**0.342**	0.358	0.352	0.353	0.348	0.346	0.348	0.358	0.349	0.354	0.348	**0.345**	0.381	0.398	0.418	0.365	0.359	0.419	0.428
	Avg	**0.245**	**0.267**	0.254	0.284	0.253	0.277	0.249	0.280	0.258	0.279	0.259	0.281	0.265	0.317	0.259	0.315	0.259	0.287	0.338	0.382
ETTh1	96	0.390	**0.397**	0.396	0.406	0.386	0.405	0.391	0.409	0.386	0.405	0.414	0.419	0.386	0.400	0.423	0.448	**0.384**	0.402	0.449	0.459
	192	0.455	0.437	0.462	0.443	0.443	0.437	**0.436**	0.445	0.441	0.436	0.460	0.445	0.437	0.432	0.471	0.474	**0.436**	**0.429**	0.500	0.482
	336	0.503	0.464	0.511	0.472	0.489	0.468	**0.476**	**0.455**	0.487	0.458	0.501	0.466	0.481	0.459	0.570	0.546	0.491	0.469	0.521	0.496
	720	0.526	0.504	0.534	0.512	0.502	0.489	**0.475**	**0.470**	0.503	0.491	0.500	0.488	0.519	0.516	0.653	0.621	0.521	0.500	0.514	0.512
	Avg	0.469	0.450	0.476	0.458	0.455	0.450	**0.445**	**0.444**	0.454	0.447	0.469	0.454	0.456	0.452	0.529	0.522	0.458	0.450	0.496	0.487
ETTh2	96	**0.290**	**0.338**	0.295	0.344	0.296	0.348	0.293	0.347	0.297	0.349	0.302	0.348	0.333	0.387	0.745	0.584	0.340	0.374	0.346	0.388
	192	0.383	0.396	0.389	0.403	0.376	0.396	**0.373**	**0.395**	0.380	0.400	0.388	0.400	0.477	0.476	0.877	0.656	0.402	0.414	0.456	0.452
	336	0.428	0.432	0.435	0.441	0.424	0.431	**0.422**	**0.430**	0.428	0.432	0.426	0.433	0.594	0.541	1.043	0.731	0.452	0.452	0.482	0.486
	720	0.433	0.444	0.441	0.452	0.426	0.444	**0.414**	**0.442**	0.427	0.445	0.431	0.446	0.831	0.657	1.104	0.763	0.462	0.468	0.515	0.511
	Avg	0.383	**0.402**	0.390	0.410	0.381	0.405	**0.375**	0.404	0.383	0.407	0.387	0.407	0.559	0.515	0.942	0.684	0.414	0.427	0.450	0.459
ETTm1	96	**0.326**	**0.351**	0.333	0.358	0.333	0.368	0.327	0.363	0.334	0.368	0.329	0.367	0.345	0.372	0.404	0.426	0.338	0.375	0.505	0.475
	192	0.378	**0.377**	0.385	0.384	0.376	0.390	0.370	0.389	0.377	0.391	**0.367**	0.385	0.380	0.389	0.450	0.451	0.374	0.387	0.553	0.496
	336	0.408	**0.399**	0.415	0.406	0.408	0.413	0.405	0.415	0.426	0.420	**0.399**	0.410	0.413	0.413	0.532	0.515	0.410	0.411	0.621	0.537
	720	0.472	**0.434**	0.481	0.442	0.475	0.448	0.459	0.450	0.491	0.459	**0.454**	0.439	0.474	0.453	0.666	0.589	0.478	0.450	0.671	0.561
	Avg	0.396	**0.390**	0.403	0.397	0.398	0.405	0.390	0.405	0.407	0.410	**0.387**	0.400	0.403	0.407	0.513	0.496	0.400	0.406	0.588	0.517
ETTm2	96	**0.175**	**0.253**	0.181	0.260	0.179	0.263	0.177	0.266	0.180	0.264	**0.175**	0.259	0.193	0.292	0.287	0.366	0.187	0.267	0.255	0.339
	192	**0.241**	**0.297**	0.248	0.305	0.250	0.309	0.245	0.306	0.250	0.309	**0.241**	0.302	0.284	0.362	0.414	0.492	0.249	0.309	0.281	0.340
	336	**0.300**	**0.334**	0.308	0.342	0.312	0.349	0.305	0.344	0.311	0.348	0.305	0.343	0.369	0.427	0.597	0.542	0.321	0.351	0.339	0.372
	720	0.401	**0.394**	0.411	0.402	0.411	0.406	**0.399**	0.402	0.412	0.407	0.402	0.400	0.554	0.522	1.730	1.042	0.408	0.403	0.433	0.432
	Avg	**0.279**	**0.319**	0.287	0.327	0.288	0.332	0.282	0.331	0.288	0.332	0.281	0.326	0.350	0.401	0.757	0.610	0.291	0.333	0.327	0.371
1st Count		**44**		0		0		15		7		12		3		1		3		0	
Avg 1st Count		**11**		0		0		3		1		1		0		0		0		0	

Table 4: Performance comparison under irregular exogenous perturbations on ETTm2 ($T = 96$). The Δ indicates the relative performance degradation (higher Δ means worse robustness) compared to the pristine (Original) setting.

Models	**Original (Pristine)**		**Mask-30% (Missing)**			**Shift+2 (Misaligned)**		
	MSE	MAE	MSE	MAE	Δ MSE	MSE	MAE	Δ MSE
iTransformer	0.180	0.264	0.207	0.291	+15.0%	0.221	0.301	+22.8%
TimeXer	0.177	0.266	0.197	0.286	+11.3%	0.204	0.291	+15.3%
CTIA-ODE (Ours)	**0.175**	**0.253**	**0.186**	**0.266**	**+6.3%**	**0.194**	**0.267**	**+10.9%**

Table 6: Sensitivity analysis of numerical ODE solvers ($T = 720$).

ODE Solver	**Weather**		**ETTm2**	
	MSE	MAE	MSE	MAE
Euler (1st-order)	0.366	0.354	0.415	0.410
Midpoint (2nd-order)	0.358	0.347	0.407	0.401
RK4 (4th-order)	**0.351**	**0.342**	**0.401**	**0.394**

Table 7: Sensitivity analysis of ORHL hyperparameters (δ and λ) on the Weather dataset. The optimal combination is highlighted in bold.

		λ (Gradient Saturation)		
	MSE / MAE	1.0	**5.0**	10.0
δ (Threshold)	0.5	0.268 / 0.293	0.265 / 0.284	0.270 / 0.289
	1.0	0.261 / 0.285	**0.245 / 0.267**	0.256 / 0.277
	2.0	0.258 / 0.281	0.265 / 0.282	0.267 / 0.297

Table 8: ODE integration versus discrete recurrent attention on Weather ($L = 96$, $T = 96$). Values are means $\pm$ sample standard deviations across three paired seeds (2026, 2027, 2028).

Model variant	Test MSE ↓	Test MAE ↓	Parameters	Latency (ms) ↓
Discrete recurrent attention	0.15653 ± 0.00059	0.19461 ± 0.00088	514,508	30.50 ± 0.14
CTIA-ODE	$\mathbf{0.15496 \pm 0.00066}$	$\mathbf{0.19359 \pm 0.00066}$	514,637	36.30 ± 0.08

Table 9: Comparison of scaled Huber loss and ORHL on Weather ($L = 96$, $T = 720$, seed=2026). Errors are displayed to five decimal places.

Objective	Scaling coefficient (c)	Test MSE ↓	Test MAE ↓
Huber ($\delta = 1$)	1	0.35488	0.34816
	2	0.35475	0.34814
	3	0.35502	0.34832
	4	0.35510	0.34837
	5	0.35510	0.34837
ORHL (Ours)	–	**0.35180**	**0.34242**

Table 10: Cross-architecture loss comparison on Weather ($L = 96$, $T = 96$, seed=2026). The CTIA-ODE row just reports its previously obtained result for this setting.

Architecture	Training loss	Test MSE ↓	Test MAE ↓	Improvement (MSE / MAE)
DLinear	MSE	0.196	0.255	–
	ORHL	0.190	0.242	3.07% / 5.21%
iTransformer	MSE	0.174	0.214	–
	ORHL	0.166	0.201	4.31% / 5.81%
CTIA-ODE	ORHL	**0.153**	**0.191**	–

Table 11: Width sensitivity and inference resource trade-offs on Weather ($L = 96$, $T = 96$, seed=2026).

Model	Width	Test MSE ↓	Test MAE ↓	Parameters	Latency (ms/batch) ↓	Peak memory (MiB) ↓
TimeXer	128	0.165	0.214	368,096	24.46	224.04
TimeXer	512	0.165	0.214	4,617,824	171.06	882.00
CTIA-ODE	128	**0.153**	**0.191**	514,637	38.24	67.16
CTIA-ODE	512	0.158	0.196	7,310,669	220.97	258.19

Table 12: Impact of Loss Function Selection and Norm-5 Gradient Clipping on Weather Benchmark ($L = 96$, averaged over $T \in \{96, 192, 336, 720\}$). Max $\|\mathbf{g}\|_2$ denotes the maximum global L2 gradient norm observed across all training steps before clipping.

Loss Objective	**w/ Norm-5 Clipping**		**w/o Clipping (Unconstrained)**		**Max $\|\mathbf{g}\|_2$**	**Clip Trigger Rate**
	MSE	MAE	MSE	MAE		
MSE Loss	0.250	0.277	0.250	0.277	0.781	0%
MAE Loss	0.248	0.270	0.248	0.270	0.375	0%
ORHL (Ours)	**0.245**	**0.267**	**0.245**	**0.267**	2.060	0%

Table 13: Forecasting performance (MSE / MAE) averaged over prediction horizons $T \in \{96, 192, 336, 720\}$ for varying look-back window sizes (L) on the Weather and ETTm1 datasets.

Dataset	**Model**	$L = 96$	$L = 192$	$L = 336$	$L = 720$
Weather	Autoformer	0.338 / 0.382	0.325 / 0.370	0.312 / 0.358	OOM
	DLinear	0.265 / 0.317	0.253 / 0.306	0.245 / 0.297	0.239 / 0.290
	CTIA-ODE	**0.245 / 0.267**	**0.242 / 0.268**	**0.236 / 0.264**	**0.237 / 0.270**
ETTm1	Autoformer	0.588 / 0.517	0.565 / 0.498	0.542 / 0.480	OOM
	DLinear	0.403 / 0.407	0.385 / 0.392	0.377 / 0.389	0.376 / 0.380
	CTIA-ODE	**0.396 / 0.390**	**0.369 / 0.378**	**0.361 / 0.376**	**0.360 / 0.380**